

Boiler Feed Pump - Operation & Maintenance Manual

Barrel-type multistage centrifugal BFP, 50% MCR, motor-driven

Field	Value
Document No.	BFP-OM-4471
Revision	Rev 3.1
Equipment OEM	(synthetic; KSB/BHEL class)
Applicable standard	ISO 10816 / IEC / CEA norms
Classification	O&M reference - SYNTHETIC (Ballast demo)
Owner	Plant O&M / Reliability

Disclaimer: This is a synthetic document generated for the Ballast demo. Values are grounded in public standards and typical industry practice but do not represent any specific OEM's proprietary manual.

1. Equipment Overview

The Boiler Feed Pump (BFP) delivers feedwater from the deaerator to the boiler drum against full boiler pressure. Units are configured 3x50% (two running, one standby). Loss of a running BFP without an available standby forces a unit derating; sustained loss forces a unit trip. BFP is therefore a Criticality 'A' asset.

Parameter	Rating
Type	Barrel (double casing) multistage centrifugal
Capacity	50% MCR (~ 620 t/h at 500 MW)
Rated head	~ 2050 m (approx 200 bar discharge)
Drive	HT motor, ~ 9,500 kW
Speed	~ 5,200 rpm via hydraulic coupling
Bearings	DE tilting-pad thrust + journal; NDE journal
Lube oil	ISO VG 46 turbine oil, forced lubrication

2. Vibration Limits (ISO 10816-3, large machines, flexible support)

Overall velocity RMS measured at bearing housings (DE/NDE, horizontal + vertical). The alarm and trip philosophy below is applied in the DCS/CBM system.

Zone	Vel. RMS (support-A)	Vel. RMS (support-B)	Action
Zone A (new/good)	<= 2.3	<= 3.5	Newly commissioned; ideal
Zone B (acceptable)	2.3 - 4.5	3.5 - 7.1	Unrestricted long-term operation
Zone C (alarm)	4.5 - 7.1	7.1 - 11.0	Short-term only; plan corrective action
Zone D (danger)	> 7.1	> 11.0	Risk of damage; trip / immediate action

Applied set-points for this BFP

Set-point	Value	Response
ALERT (Zone C entry)	4.5 mm/s	Raise CBM alert; increase monitoring; plan action
DANGER (Zone D entry)	7.1 mm/s	Prepare controlled derate; ready standby
TRIP (sustained)	> 7.1 mm/s for > 10 min	Trip pump; transfer to standby

Interpretation note: a rising trend from Zone B into Zone C is more significant than an absolute reading. A DE-bearing vibration climbing over days typically indicates thrust-bearing wear or shaft misalignment (see Section 5).

3. Bearing Temperature Limits

Point	Normal	Alarm	Trip
DE thrust bearing metal temp	65 - 85 degC	95 degC	105 degC
NDE journal bearing metal temp	60 - 80 degC	90 degC	100 degC
Lube oil supply temp	38 - 45 degC	50 degC	55 degC

4. Preventive Maintenance & Service Intervals

Task	Interval	Notes
Lube oil sampling (ISO 4406)	Monthly	Trend particle count; target <= 18/16/13
Vibration route survey	Weekly	Handheld backup to online CBM

DE thrust bearing inspection/replace	8,000 running hrs	Or on vibration/temp trend breach
Mechanical seal replacement	16,000 running hrs	Or on seal leak-off rise
Coupling alignment check	Annual / after overhaul	Laser alignment
Full overhaul	48,000 running hrs	Rotor, bearings, seals, clearances

MTBF (design): 30,000 - 45,000 hours. Critical insurance spare: DE thrust bearing assembly (long procurement lead time - keep on stock).

5. Common Failure Modes (ISO 14224) & Root Cause

Failure mode	Mechanism	Early indicator	Typical MTTR
Bearing failure	Fatigue / lube contamination	Rising DE vibration + bearing temp	72 h
Vibration high	Misalignment / imbalance	1x/2x running-speed vib rise	48 h
Seal leakage	Wear / dry-run	Seal leak-off flow rise	24 h
Cavitation	Low suction / NPSH	Broadband noise, flow instability	Variable

Bearing-failure playbook: when DE vibration trends from Zone B toward Zone C/D over several days with a coincident bearing-temperature rise, the thrust bearing is the prime suspect. Confirm with oil-particle trend. If no healthy standby is available and the critical bearing spare is not in stock, pre-emptively derate the unit to protect the grid commitment rather than risk an uncontrolled trip.